

NORTH SYDNEY GIRLS HIGH SCHOOL

Year 12

HSC Mathematics Extension 1 Assessment Task 2

Term 1 2008

Name: _____ Mathematics Class: _____

Time Allowed: 60 minutes + 2 minutes reading time

Available Marks: 55

Instructions:

- Start each question on a new page.
- Attempt all six questions.
- Show all necessary working.
- Marks may be deducted for incomplete or poorly arranged work.
- Write on one side of the page only. Do not work in columns.
- Questions are of approximately equal value.

Question	1	2	3a	3b	4	5a	5b	6	Total
PE6	/9	/8		/5	/10	/5		/11	/48
HE2			/4				/3		/7
									/55
									%

Question 1: (9 marks)**Marks**

- a) Evaluate $\sum_{r=1}^3 (-1)^{r+1} r^2$ **1**
- b) What is the least integer value of n for which $1 + 3 + 3^2 + \dots + 3^{n-1} > 10^4$? **3**
- c) (i) Find the n^{th} term of each of the series below **2**
 $A_n = 3 + 6 + 12 + \dots$
 $B_n = 2 + 7 + 12 + \dots$
- (ii) Deduce from the terms of the series, A_n and B_n , the fourth term of the following series, C_n where **1**
 $C_n = 10 + 26 + 48 + \dots$
- (iii) What is the n^{th} term of the series C_n ? **2**

Question 2: (8 marks) Start a new page

- a) For what values of x is the graph of the function $f(x) = 2x^3 - 6x$ both concave upwards and decreasing? **2**
- b) Ada borrows \$240 000 to purchase a house. A compound interest rate of 6% per annum is calculated on the balance of the loan at the end of each month. Equal monthly repayments of \$ W are made at the end of each month, immediately after the interest calculation. The loan is to be repaid over 20 years.
- (i) Show that A_2 , the amount owing on the loan after 2 months is given by **1**
 $A_2 = 240000(1.005)^2 - W(1 + 1.005)$
- (ii) Deduce a similar expression for the amount owing after 20 years **1**
- (iii) Show that **2**
$$W = \frac{1200 \times (1.005)^{240}}{(1.005)^{240} - 1}$$
- (iv) When does the balance owing first fall below \$200 000? **2**
Answer correct to the nearest month.

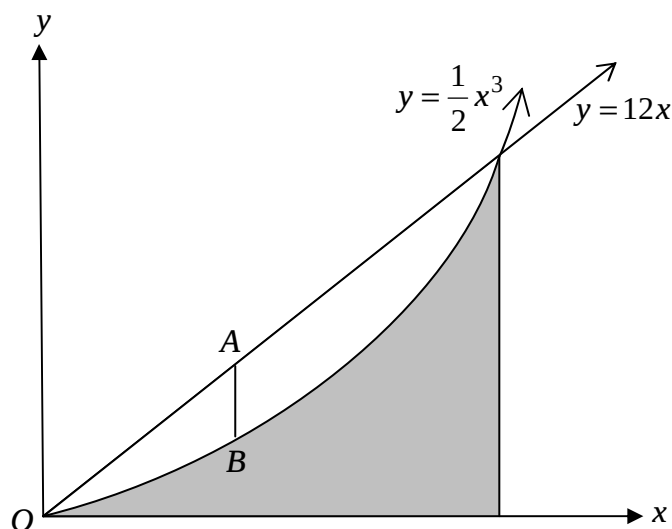
Question 3: (9 marks) Start a new page**Marks**

- a) Use the Mathematical induction to prove that 4
 $\sin(x + n\pi) = (-1)^n \sin x$ for $n = 1, 2, 3, \dots$
- b) Students at International High School must study at least one of the two languages, English and Mandarin. At a meeting of 28 students from the school, 18 study English and 22 study Mandarin.
- (i) Draw a Venn (or similar) diagram illustrating this information. 2
- (ii) Hence find the probability that at the meeting:
- (α) one randomly selected student studies the subject of English. 1
- (β) two randomly selected students both study the subject of English. 1
- (γ) one randomly selected student studies both the languages specified. 1

Question 4: (10 marks) **Start a new page**

- a) A linear pipe is placed above a ski slope as shown below. The pipe and the cable are defined by the equations $y = 12x$ and $y = \frac{1}{2}x^3$ respectively.

Vertical supports for the pipe are constructed along the slope under the pipe as illustrated in the diagram by the vertical support AB . Dimensions are in metres.



- (i) Show that the height h of each support is given by $h = \frac{x}{2}(24 - x^2)$ 2
- (ii) Find the height of the tallest vertical support that can be placed between the pipe and the ski slope within the context of the diagram. 4

Question 4 continues

Question 4 continued:**Marks**

- b) Tom contributes to a superannuation fund. At the start of every quarter (of a year), he contributes \$250. The investment pays interest at 8% per annum compounded quarterly. This contribution continues for 30 years.
- (i) What amount does Tom contribute altogether? **1**
 - (ii) How much does the initial contribution of \$250 reach at the end of thirty years? **1**
 - (iii) Find the total value of Tom's fund after thirty years. **2**

Question 5: (8 marks) Start a new page

- a) Jenny devises a game of chance for one person and plays it herself. She throws two unbiased dice repeatedly until the sum of the numbers displayed is either 9 or 12. If the sum is 9, Jenny wins the game. If the sum is 12, Jenny loses the game. If the sum is any other number, then Jenny throws again.
- (i) Show that the probability that Jenny wins on the first throw is $\frac{1}{9}$. **1**
 - (ii) Show that the probability that a game continues with Jenny winning on the second throw is given by $\frac{1}{9} \times \frac{31}{36}$. **2**
 - (iii) Use your knowledge of series to find the probability that Jenny will eventually win the game. **2**
- b) Use Mathematical Induction to prove that for integers $n \geq 1$, $9^{n+2} - 4^n$ is a multiple of 5. **3**

- a) A continuous function $y = f(x)$ has its second derivative defined by 2

$$f''(x) = -\frac{4x(3-x^2)}{(1+x^2)^3}$$

The function $f(x)$ has three points of inflexion. Two of these points are at $(\sqrt{3}, \frac{\sqrt{3}}{2})$ and $(-\sqrt{3}, -\frac{\sqrt{3}}{2})$. Show that there is a third point of inflexion.

- b) Consider the function $y = \frac{2x}{x^2+1}$.

- (i) Show that this function has two stationary points. 3

- (ii) This function has its second derivative defined by 3

$$f''(x) = -\frac{4x(3-x^2)}{(1+x^2)^3}$$

Hence, or otherwise, classify the stationary points found in part (i).

- (iii) Using the information in both part a) and b) (i) and (ii), sketch the 3
graph of $y = \frac{2x}{x^2+1}$ showing all of its important features including those found above.

End of Paper

Question 1

a) $\sum_{r=1}^9 (-1)^{r+1} r^2 = 1 - 4 + 9$
 $= 6$

b) $S_n = 1 \left(\frac{3^n - 1}{2} \right)$

$\frac{3^n - 1}{2} > 10^4$

$3^n > 2 \cdot 10^4 + 1$

$3^n > 20001$

$n > 9$ [Trial/ Error]

$n = 10$

c) (i) $A_n: T_n = 3 \cdot 2^{n-1}$

$B_n: T_n = 2 + (n-1)5$
 $= 5n - 3$

(ii) $T_4 = 82$

(iii) $T_n = 2(3 \cdot 2^{n-1} + 5n - 3)$
 $= 3 \cdot 2^n + 10n - 6$

Question 2

a) $f'(x) = 6x^2 - 6$

$f''(x) = 12x$

Decreasing function:

$f'(x) \leq 0$

$6(x+1)(x-1) \leq 0$

$-1 \leq x \leq 1$ ①

Concave upward:

$f''(x) > 0$ i.e. $x > 0$ ②

Solving ①, ② simultaneously

$0 \leq x \leq 1$

QUESTION 1b)

2b) i) Let A_n be amount owing after n months

(i) $A_1 = 240000(1.005) - W$

$\therefore A_2 = A_1(1.005) - W$
 $= [240000(1.005) - W]1.005 - W$
 $= 240000(1.005)^2 - W(1 + 1.005)$

(ii) $A_{240} = 240000(1.005)^{240} - W(1 + 1.005 + \dots + 1.005^{239})$

(iii) Let $A_{240} = 0$

$W \left(\frac{1.005^{240} - 1}{0.005} \right) = 240000(1.005)^{240}$

$\therefore W = \frac{1200(1.005)^{240}}{(1.005)^{240} - 1}$

iv) $A < 200000$

Now

$A_n = 240000(1.005)^n - \frac{W((1.005)^n - 1)}{0.005}$

We know that

$W = 1719.43$

So $240000(1.005)^n - \frac{1719((1.005)^n - 1)}{0.005} < 200000$

$1200(1.005)^n - 1719(1.005)^n + 1719 < 100000$

$(1.005)^n (1200 - 1719) < -719$

$519(1.005)^n > 719$

$(1.005)^n > \frac{719}{519}$ (1.3)

By calculator or logarithms

$n > 65.3$

i.e. $n = 66$

66 months

QUESTION 3

a) $\sin(x+n\pi) = (-1)^n \sin x$

Step 1

Test for $n=1$

$$\text{LHS} = \sin(x+\pi)$$

$$\text{RHS} = -\sin x$$

$$\text{LHS} = \sin(x+\pi) \stackrel{0}{=} \sin x \cos \pi + \cos x \sin \pi$$

$$= -\sin x$$

$$= \text{RHS}$$

Step 2

Assume that

$$\sin(x+k\pi) = (-1)^k \sin x$$

and hence show that

$$\sin(x+(k+1)\pi) = (-1)^{k+1} \sin x$$

$$\text{LHS} = \sin(x+(k+1)\pi)$$

$$= \sin((x+k\pi)+\pi)$$

$$= \sin(x+k\pi) \cos \pi + \cos(x+k\pi) \sin \pi$$

$$= -\sin(x+k\pi)$$

$$= (-1)^{k+1} \sin x \quad [\text{By assumption}]$$

$$= (-1)^{k+1} \sin x$$

$$= \text{RHS}$$

Step 3

Since the result is

true for $n=1$ and is

true for $n=k+1$ if

true for $n=k$ then it

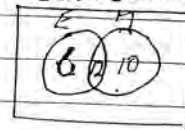
is true for $n=2$

and so on for $n=3, 4, 5, \dots$

b)

MEETING OF 28

(1)

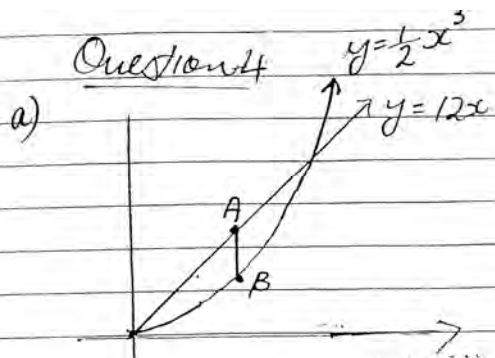


$$(1) (2) \frac{9}{14}$$

$$(3) \frac{18 \times 17}{28 \times 27} = \frac{1}{2}$$

$$B \frac{3}{7}$$

Question 4



Let A be (x, y) ie $(x, 12x)$
 B be (x, y) ie $(x, \frac{1}{2}x^3)$

$$AB = 12x - \frac{1}{2}x^3$$

(i) i.e. $h = \frac{1}{2}x(24 - x^2)$

Q4b)

Find $\frac{dh}{dx}$

68) \$30000

(ii) $250(1.02)^{20}$

\$2691.29

(ii) $\frac{dh}{dx} = 12 - \frac{3}{2}x^2$

Put $\frac{dh}{dx} = 0$

(iii)

$V = 250(1.02 + (1.02)^2 + (1.02)^3 + \dots + (1.02)^{20})$

$= 250 \frac{1.02((1.02)^{20} - 1)}{0.02}$

$\frac{3x^2}{2} = 12$

$3x^2 = 24$

$x = 2\sqrt{2}$

\$124505.83₍₂₎

$\therefore h = \frac{2\sqrt{2}}{2} (24 - (2\sqrt{2})^3)$
 $= \sqrt{2} (24 - 8)$
 $= 16\sqrt{2}$

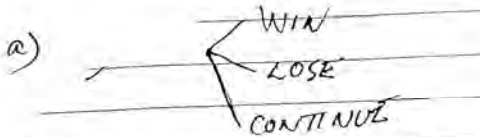
Check Maximum

$\frac{d^2h}{dx^2} = -3x$

$\frac{d^2h}{dx^2} = -48\sqrt{2} < 0$

implies at
 maximum
 value

QUESTIONS



SUM OF 9: $(4,5) (5,4) (3,6) (6,3)$

$$P(\text{SUM OF 9}) = \frac{1}{9}$$

$$P(\text{SUM OF 12}) = \frac{1}{36}$$

$$P(\text{ANOTHER SUM}) = \frac{31}{36}$$

(i) $P(W) = \frac{1}{9}$

(ii) $P(\text{ANOTHER THEN SUM OF 9})$

$$\text{ie } P(\text{ANOTHER SUM}) \cdot P(\text{SUM OF 9}) = \frac{31}{36} \cdot \frac{1}{9}$$

(iii) $= \frac{1}{9} \cdot \frac{31}{36}$

THROW 1: THROW 2: ... THROW 3:

$$\frac{31}{36} \cdot \frac{1}{9} \quad \frac{31}{36} \cdot \frac{31}{36} \cdot \frac{1}{9} \quad \frac{31}{36} \cdot \frac{31}{36} \cdot \frac{1}{9}$$

$$\text{ie } S = \frac{1}{9} + \frac{1}{9} \cdot \frac{31}{36} + \frac{1}{9} \cdot \frac{31}{36} \cdot \frac{31}{36}$$

$$= \frac{1}{9}$$

$$1 - \frac{31}{36}$$

$$= \frac{5}{36} = \frac{4}{5}$$

b) Step 1:

For $n=1$, $9^3 - 4 = 725$ which is a multiple of 5

Step 2:

Assume the formula true for $n=k$ ie $9^{k+2} - 4 = 5N$ and show that it follows true for $n=k+1$ ie show

$$9^{k+3} - 4 = 5M \quad (N, M \text{ INTEGERS})$$

$$\text{LHS} = 9^{k+3} - 4$$

$$= 9 \cdot 9^{k+2} - 4 \cdot 4^k$$

$$= 9(5N + 4^k) - 4 \cdot 4^k$$

$$= 45N + 5 \cdot 4^k$$

$$= 5(9N + 4^k)$$

$$= 5M$$

$$= \text{RHS}$$

Step 3 Since the result is

true for $n=1$ and is true

for $n=k+1$ if true for $n=k$,

then it is true for $n=2$

and so on for all $n=3, 4, \dots$

Question: 6

$$a) f''(x) = -\frac{4x(3-x^2)}{(1+x^2)^3}$$

$$f''(0) = 0$$

$$\text{AND } f''(-1) = \frac{4(2)}{8} > 0 \Rightarrow \text{CONC UP}$$

$$f''(1) = -\frac{4(2)}{8} < 0 \Rightarrow \text{CONCAVE DOWNWARD}$$

$(0,0)$ is a point of inflexion

$$b) y = \frac{2x}{x^2+1}$$

$$y' = \frac{2(x^2+1) - 2x \cdot 2x}{(x^2+1)^2}$$

$$= \frac{2 - 2x^2}{(x^2+1)^2}$$

Put $y' = 0$ for stationary points

$$\frac{2 - 2x^2}{(x^2+1)^2} = 0$$

$$(x^2+1)^2$$

$$1 - x^2 = 0$$

$$x = \pm 1$$

$(1,1)$ and $(-1,-1)$

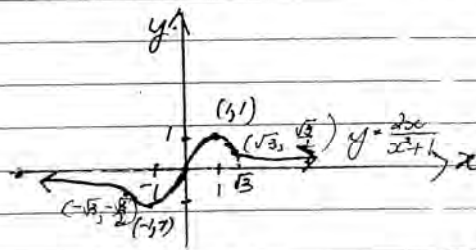
Test $x=1$

$$f''(1) = -\frac{4 \cdot 2}{8} < 0 \Rightarrow \text{CONCAVE DOWN}$$

$$f''(-1) = -\frac{4 \cdot 2}{8} > 0 \Rightarrow \text{CONCAVE UPWARD}$$

$(-1,-1)$ is a minimum turning point

$(1,1)$ is a maximum turning point



Asymptote

$$y = \lim_{x \rightarrow \infty} \frac{2x}{x^2+1}$$

$$\text{i.e. } y = 0$$

Extra Detailed Solutions

(2) (b) Let A_n be the amount owing after n months.

$$(i) \quad A_1 = 240000(1.005) - W$$

$$\begin{aligned}\therefore A_2 &= A_1(1.005) - W \\ &= [240000(1.005) - W](1.005) - W \\ &= 240000(1.005)^2 - W(1 + 1.005)\end{aligned}$$

$$(ii) \quad A_{240} = 240000(1.005)^{240} - W(1 + 1.005 + \dots + 1.005^{239})$$

(iii) Let $A_{240} = 0$

$$\therefore 240000(1.005)^{240} - W(1 + 1.005 + \dots + 1.005^{239}) = 0$$

$$\therefore \frac{W(1.005^{240} - 1)}{0.005} = 240000(1.005)^{240}$$

$$\therefore W = \frac{1200(1.005)^{240}}{1.005^{240} - 1}$$

$$\therefore W = 1719.43$$

(iv) When is $A_n < 200\,000$

$$\text{Now } A_n = 240000(1.005)^n - \frac{W(1.005^n - 1)}{0.005}$$

We know that $W = 1719.43$

$$\therefore 240\,000(1.005)^n - \frac{1719.43(1.005^n - 1)}{0.005} < 200\,000$$

$$\therefore 1200(1.005)^n - 1719.43(1.005^n) + 1719.43 < 1000$$

$$\therefore 1.005^n(1200 - 1719.43) < -719.43$$

$$\therefore 519(1.005^n) > 719.43$$

$$\therefore 1.005^n > \frac{719.43}{519}$$

$$\therefore n > 65.3$$

So 66 months

ALTERNATIVE PROBLEM for 3(a)

(a) $\cos(x + n\pi) = (-1)^n \cos x$

Test $n = 1$:

$$\begin{aligned}
 \text{LHS} &= \cos(x + \pi) \\
 &= \cos(\pi + x) \\
 &= -\cos x && \left[\text{angles of any magnitude, 3rd quadrant} \right] \\
 &= (-1)^1 \cos x \\
 &= \text{RHS}
 \end{aligned}$$

So it is true for $n = 1$.

Assume true for $n = k$, ie $\cos(x + k\pi) = (-1)^k \cos x$ (1)

NTP true for $n = k + 1$, ie $\cos[x + (k + 1)\pi] = (-1)^{k+1} \cos x$

$$\begin{aligned}
 \cos[x + (k + 1)\pi] &= \cos[\pi + (x + k\pi)] \\
 &= -\cos(x + k\pi) && \left[\text{angles of any magnitude} \right] \\
 &= -[(-1)^k \cos x] && \left[\text{from (1)} \right] \\
 &= (-1)^{k+1} \cos x
 \end{aligned}$$

So if $n = k$ is true then it is true for $n = k + 1$ and so by the principle of mathematical induction it is true for all $n \geq 1$

(4) (b) 8% pa = 2% per quarter; 30 years = 120 quarters

(i) $250 \times 120 = 30\,000$

(ii) The first \$250 is invested for 120 quarters and so accrues $250(1.02)^{120}$

(iii) The next \$250 is invested for 119 quarters and so accrues $250(1.02)^{119}$, and so on until the last \$250 accrues $250(1.02)$.

The total lump sum, \$ L is given by:

$$\begin{aligned}
 L &= 250(1.02)^{120} + 250(1.02)^{119} + \dots + 250(1.02) \\
 &= 250[1.02 + \dots + 1.02^{120}] \\
 &= 250 \times S_{120} && [a = 1.02, r = 1.02] \\
 &= 250 \times \frac{1.02(1.02^{120} - 1)}{1.02 - 1} \\
 &= 250 \times 51(1.02^{120} - 1) \\
 &\approx 124\,505.83
 \end{aligned}$$

Tom earns \$124 505.83

Question 4:

a) (i) $AB = h = 12x - \frac{1}{2}x^3$ ① $A = (x, 12x)$ & $B = (x, \frac{1}{2}x^3)$
 $= \frac{x}{2}(24 - x^2)$ ①

(iv) $\frac{dh}{dx} = 12 - \frac{3}{2}x^2$ } ①
 $\frac{d^2h}{dx^2} = -3x$

at max/min, $\frac{dh}{dx} = 0 \therefore 12 - \frac{3x^2}{2} = 0$

$\frac{3x^2}{2} = 12$

$x^2 = 8$ ①

$x = \sqrt{8}$ as $x > 0$ ①

then $\frac{d^2h}{dx^2} = -3(\sqrt{8})$

$\frac{d^2h}{dx^2} < 0 \Rightarrow$ a maximum occurs ①

$\therefore$ max height $= 12\sqrt{8} - \frac{1}{2}(\sqrt{8})^3$ m
 $= 12\sqrt{8} - 4\sqrt{8}$ m
 $= 8\sqrt{8}$ m ①

(b) (i) Total contribution $= 4 \times \$250 \times 30$
 $= \$30000$ ①

(ii) Initial contribution grows to $\$(250)(1.02)^{120}$ ①
 $= \$2691.29$

(iii) second contribution grows to $\$(250)(1.02)^{119}$ ①
 Third contribution grows to $\$(250)(1.02)^{118}$ ①

$\therefore$ Total $= 250(1.02) + 250(1.02)^2 + 250(1.02)^3 + \dots + 250(1.02)^{120}$
 $= \frac{a(r^n - 1)}{r - 1}$ where $a = 250(1.02)$ ①
 $r = 1.02$
 $n = 120$

$= \frac{250(1.02)(1.02^{120} - 1)}{0.02}$ ①

$= 124505.828\dots$

$\therefore$ Fund is worth $\$124505.83$ (rounded) ①